

Fraud Detection Using Machine Learning

Project Summary

This project focuses on developing an advanced fraud detection system using machine learning techniques to identify fraudulent financial transactions. The system was built using transaction and identity datasets and includes complete end-to-end data analysis, preprocessing, model training, explainable AI, risk segmentation, and dashboard deployment.

The project began with data loading, merging, exploratory data analysis, and handling missing values. Significant preprocessing steps were performed including feature engineering, label encoding, scaling, and class imbalance handling using SMOTE.

Multiple machine learning models including LightGBM, XGBoost, and Isolation Forest were trained and evaluated using metrics such as Precision, Recall, F1-Score, ROC-AUC, and PR-AUC. XGBoost achieved the best overall performance for fraud detection.

Explainable AI was implemented using SHAP values to understand feature importance and explain individual fraud predictions. Risk segmentation techniques were also used to classify transactions into Critical Risk, Suspicious, and Clear categories based on fraud probability.

An interactive multi-page Streamlit dashboard was developed to visualize fraud analytics, transaction exploration, SHAP explanations, and risk insights in real time.

The project demonstrates the practical application of machine learning and explainable AI in financial fraud detection systems and provides valuable business insights for fraud prevention and operational monitoring.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- LightGBM
- SHAP
- Streamlit

- Plotly
 - Matplotlib
 - Seaborn
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Key Features

- Fraud transaction prediction
 - Class imbalance handling using SMOTE
 - SHAP explainability analysis
 - Risk segmentation
 - Interactive Streamlit dashboard
 - Visual fraud analytics
 - Threshold optimization
 - Business recommendations for fraud prevention
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Conclusion

The developed fraud detection system successfully identifies fraudulent transactions with strong predictive performance and interpretable results. The project highlights how machine learning and explainable AI can support real-world fraud monitoring and financial risk management systems.